

Capture 3. 马尔可夫链.

例 3.5.5. 根据题意, 做类似吸收态的处理.

习题:

$$1. \begin{array}{cc} A & B \\ m & m \\ \text{自} & x_0 \quad m-x_0 \\ \text{黑} & m-x_0 \quad x_0 \end{array} \quad \begin{array}{l} I = \{0, 1, 2, \dots, m\} \\ P_{i,i+1} = \frac{m-i}{m} \cdot \frac{m-i}{m} \\ P_{i,i} = 2 \frac{m-i}{m} \cdot \frac{1}{m} \\ P_{i,i-1} = \frac{1}{m} \cdot \frac{1}{m} \end{array}$$

$$2. S_n = x_1 + x_2 + \dots + x_n.$$

$$I = \{0, 1, 2, \dots\}.$$

$$\text{对于一步转移概率: } P_{ij} = \begin{cases} 0, & j < i \\ p_k, & j \geq i \\ & k = j - i \end{cases}$$

$$3. \{L_n\} = \{0, 1, 2, 3, \dots\}.$$

$$P_{i,i+1} = p.$$

$$P_{i0} = 1-p.$$

$$4. x_n = \{0, 1, 2, \dots, N\}.$$

$$P_{i,i+1} = 2 \frac{1}{N} \cdot \frac{N-i}{N-1} \cdot p.$$

$$P_{i,i} = 1 - 2 \frac{1}{N} \cdot \frac{N-i}{N-1} \cdot p.$$

$$5. (1) P(Y_2=1 | Y_0=1, Y_1=6) = 0$$

$$P(Y_2=1 | Y_1=6) = \frac{1}{6} \cdot \frac{1}{11} = \frac{1}{66}$$

$$(2) P(Z_2=12 | Z_0=2, Z_1=7) = P(X_4=6) = \frac{1}{6}$$

$$P(Z_2=12 | Z_1=7) = P(X_4=6) \cdot P(X_3=6 | X_1+X_2=7) = \frac{1}{26}$$

(3) 不具有.

$$6. (1) P(X_2=4) = P(X_0=2) \cdot p^2 = \frac{1}{2} p^2.$$

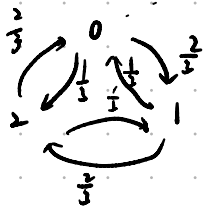
$$P(X_2=4 | X_0=0) = 0. \text{ 不相等.}$$

$$(2) P(|X_2|=2 | |X_0|=2, |X_1|=1) = p.$$

$$P(|X_2|=2 | |X_0|=0, |X_1|=1) = p^2 + (1-p)^2$$

(3) 由(2)显然, 不具有

7.4)

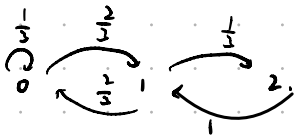


$$\begin{pmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{3} & \frac{2}{3} & 0 \end{pmatrix}$$

(2) $P(X_0=0, X_2=0, X_4=1) = \frac{1}{3} \times \frac{2}{3} \times 2 \times \frac{1}{3} \times \frac{1}{3} = \frac{2}{81}$

$P(X_2=1) = \frac{1}{2} \times \frac{1}{3} \times \frac{1}{3} + \frac{1}{4} \times \frac{2}{3} \times 2 + \frac{1}{4} \times \frac{2}{3} \times \frac{2}{3} = \frac{5}{18}$

8.



(1) $P(X_0=0|X_0=0) = \frac{1}{3} \times \frac{1}{3} + \frac{2}{3} \times \frac{2}{3} = \frac{5}{9}$

$P(X_2=0) = \frac{5}{9} + \frac{2}{3} \times \frac{1}{3} + \frac{2}{3} \times \frac{1}{3} = \frac{13}{27}$

$P(X_0=0|X_2=0) = \frac{5}{9} \times \frac{P(X_0=0)}{P(X_2=0)} = \frac{5}{13}$

(2) $P(X_1=0) = \frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{3} = \frac{1}{3}$

$P(X_1=0, X_2=0, X_4=1, X_0=1) = P(X_1=0) \cdot P(X_2=0|X_0=0) \cdot \frac{2}{3} (\frac{1}{3} + \frac{1}{3} \times \frac{2}{3}) = \frac{70}{729}$

(3) $f_{11}^{(1)} = 0, f_{11}^{(2)} = \frac{2}{9}$

$f_{11}^{(n)} = \frac{2}{9} \times \frac{2}{9} \times (\frac{1}{3})^{n-2} = \frac{4}{9} (\frac{1}{3})^{n-2}$

$f_{11} = \sum_{n=1}^{\infty} f_{11}^{(n)} = 1 \quad (n \geq 3)$

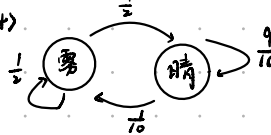
$\mu_1 = \sum_{n=1}^{\infty} n f_{11}^{(n)} = \frac{7}{3}$

9. (1) $P = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{10} = \frac{3}{10}$

(2) $\frac{1}{10} \times \frac{1}{2} = \frac{1}{20}$

(3) $P(\text{第1次=无雾霾} | \text{第2次=无雾霾}) = 0.5 \times \frac{0.5}{\frac{1}{10}} = \frac{5}{6}$

(4)



则平稳分布

$$\begin{cases} \pi_1 = \frac{1}{2}\pi_1 + \frac{1}{10}\pi_2 \\ \pi_2 = \frac{9}{10}\pi_2 + \frac{1}{2}\pi_1 \\ \pi_1 + \pi_2 = 1 \end{cases}$$

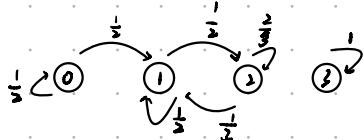
得 $\pi_1 = \frac{1}{6}, \pi_2 = \frac{5}{6}$

即雾霾天的 $\frac{1}{6}$

$\frac{1}{3}A = \frac{4}{3} + 2 = \frac{10}{3}$

+

10.



$A - \frac{1}{2}A = \frac{4}{3} - (\frac{1}{2})^1 - \dots - n(\frac{1}{2})^n$

(1) $P(X_1=1, X_3=2) = (\frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2}) \times (\frac{1}{2} \times \frac{2}{2} + \frac{1}{2} \times \frac{1}{2}) = \frac{7}{20}$

$P(X_2=1) = \frac{1}{3} (\frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2}) + \frac{1}{3} (\frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2}) = \frac{11}{18}$

$P(X_{10}=0) = \frac{1}{3} \cdot \frac{1}{20}$

(2) $\{0\}, \{1,2\}, \{3\}$. ① $f_{11} = 1$ 333常返, 平均回返时1. ② $f_{00}=0$ 暂离.

① $f_{11}^{(1)} = \frac{1}{2}, f_{11}^{(n)} = \frac{1}{6} (\frac{1}{2})^{n-2}$

$f_{11} = \sum_{n=1}^{\infty} f_{11}^{(n)} = 1, \mu_1 = \sum_{n=1}^{\infty} n f_{11}^{(n)} = \frac{5}{2}$

1,2互返, 则 $f_{12}=1$. 则 $\mu_2 = \frac{5}{2}$

(也可平稳分布 $\pi_1 = \frac{1}{2}\pi_0 + \frac{1}{2}\pi_1 + \frac{1}{2}\pi_2 \Rightarrow \frac{1}{2}\pi_1 = \frac{1}{2}\pi_2$)

11.

$$d \subset 0 \xrightarrow{1-d} d$$

$$u) f_{00}^{(0)} = d$$

$$f_{00}^{(n)} = (1-d)^2 d^{n-2}, n \geq 2.$$

$$f_{01}^{(n)} = d^{n-1} (1-d).$$

$$12. \begin{cases} x_0 = \frac{1}{3}x_0 + \frac{2}{3}x_1 \\ x_1 = \frac{2}{3}x_0 + x_2 \\ x_2 = \frac{1}{3}x_2 \\ x_1 + x_2 + x_3 = 1 \end{cases}$$

$$\text{则 } (\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$$

$$13. 1) \{x_n\} = \{0, 1, 2, 3\}$$

$$P_{01} = P_{12} = P_{23} = P_{30} = \frac{1}{6} \quad P_{02} = P_{13} = P_{20} = P_{31} = \frac{1}{6} \quad P_{03} = P_{10} = P_{21} = P_{32} = \frac{1}{6} \quad P_{00} = P_{11} = P_{22} = P_{33} = \frac{1}{6}$$

$$\text{则一步转移矩阵: } \begin{pmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \end{pmatrix}$$

$$\begin{cases} x_1 = \frac{1}{6}x_1 + \frac{1}{6}x_2 + \frac{1}{6}x_3 + \frac{1}{6}x_4 \\ x_2 = \frac{1}{6}x_1 + \frac{1}{6}x_2 + \frac{1}{6}x_3 + \frac{1}{6}x_4 \\ x_3 = \frac{1}{6}x_1 + \frac{1}{6}x_2 + \frac{1}{6}x_3 + \frac{1}{6}x_4 \\ x_4 = \frac{1}{6}x_1 + \frac{1}{6}x_2 + \frac{1}{6}x_3 + \frac{1}{6}x_4 \\ x_1 + x_2 + x_3 + x_4 = 1 \end{cases}$$

$$(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$$

$$(2) \frac{1}{4}$$

$$14. u) P_i = \frac{1}{N}P + \frac{N-1}{N}(1-P)$$

$$P_{i+1} = \frac{N-1}{N}P$$

$$P_{i+1} = \frac{1}{N}(1-P)$$

$$(2) N=3 \quad \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{3} \\ 0 & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

$$\text{则 } \vec{x}_1 = \vec{x}_1 P \quad \vec{x}_1 = (\frac{1}{6}, \frac{1}{3}, \frac{1}{3})$$

$$u_0 = \frac{1}{x_0} = 6.$$

$$15. u) \{x_n\}, \quad I = \{0, 1, 2, 3\}$$

$$P_{i+1} = P \quad P_{i0} = 1-P, \quad u \neq 0 \quad P_{03} = 1 \quad (\text{出时为0, 则到其他地方})$$

$$P = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1-P & P \\ 0 & 1-P & P & 0 \\ 1-P & P & 0 & 0 \end{pmatrix} \quad \text{则 } \vec{x}_1 = \vec{x}_1 P$$

$$\vec{x}_1 = (\frac{1-P}{4-P}, \frac{1}{4-P}, \frac{1}{4-P}, \frac{1}{4-P})$$

$$(2) \text{ 即雨天出门时 } X=0. \quad P(4-P)$$

$$\text{故 } P = \frac{P(1-P)}{4-P} \leq \frac{1}{4(4-P)} \leq \frac{1}{12}$$

$$16. \lim_{n \rightarrow \infty} P(X_n=0) = 0.$$

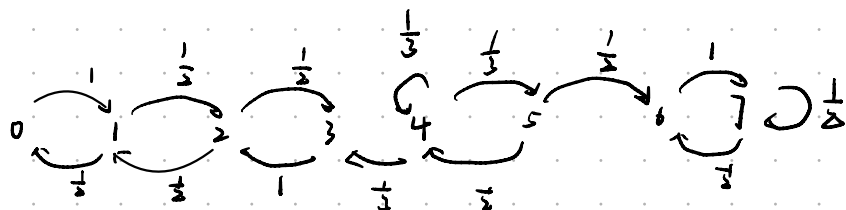
$$\lim_{n \rightarrow \infty} P(X_n=3) = \frac{1}{3}.$$

$$\begin{aligned} \lim_{n \rightarrow \infty} P(X_n=1) &= \lim_{n \rightarrow \infty} P_{01}^{(n)} P_{X_0=0} + \lim_{n \rightarrow \infty} P_{11}^{(n)} P_{X_1=1} \\ &= \frac{4}{15} \end{aligned}$$

$$\lim_{n \rightarrow \infty} P(X_n=2) = \frac{2}{5}$$

第17题.
问物物

17.



$$(1) \{0, 1, 2, 3\} \quad \{4, 5\} \quad \{6, 7\}$$

其中 $\{0, 1, 2, 3\}$ 与 $\{6, 7\}$ 是闭的.

$$(2) \text{ 易得 } d(0) = d(1) = d(2) = d(3) = 2.$$

$$d(4) = d(5) = 1$$

$$d(6) = d(7) = 1.$$

$$\text{对状态 7: } f_{77}^1 = \frac{1}{2} \quad f_{77}^2 = \frac{1}{2} \quad f_{77}^n = 0 \quad (n \geq 3)$$

则 $f_{77} = 1$ 故 $\{6, 7\}$ 常返.

$$\text{平稳分布 } (\frac{1}{3}, \frac{2}{3}) \quad \text{则 } \mu_6 = 3 \quad \mu_7 = \frac{3}{2}.$$

$$\text{对 4: } f_{44} = f_{44}^1 + f_{44}^2 = \frac{1}{6} \neq 1. \text{ 故}$$

$\{4, 5\}$ 暂离.

$$\text{对状态 3: } f_{33}^1 = 0 \quad f_{33}^2 = \frac{1}{2} \quad f_{33}^3 = 0 \quad f_{33}^n = (\frac{1}{2})^{n-2}$$

$$f_{33}^{2n} = \sum_{k=1}^{n-1} (\frac{1}{2})^k (\frac{1}{2})^{n-k} = \frac{1}{2} \cdot \frac{1}{2}^{n-1} (1 - (\frac{1}{2})^{n-1}) \quad (n \geq 2)$$

$$(k+m) = n-1$$

$$\text{则 } f_{33} = 1.$$

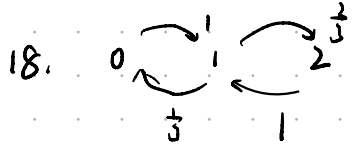
故 $\{0, 1, 2, 3\}$ 常返. 平稳分布 $(\frac{1}{6}, \frac{1}{3}, \frac{1}{3}, \frac{1}{6})$.

$$\text{则 } \mu_0 = 6 \quad \mu_1 = \mu_2 = 3 \quad \mu_3 = 6$$

$$(3) \{4, 5\} \text{ 为暂离态. 则 } \lim_{n \rightarrow \infty} P_{45}^{(n)} = 0$$

$$\lim_{n \rightarrow \infty} P_{67}^{(n)} = \pi_7 = \frac{2}{3}.$$

(4)



11) $\{0, 1, 2\}$ 闭.

对状态 1 $f_{11} = f_{11}^2 = \frac{1}{3} + \frac{2}{3} = 1$. 则 $\{0, 1, 2\}$ 常返.

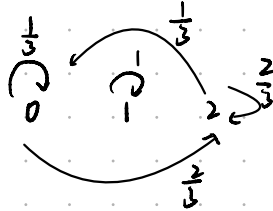
平稳分布.
$$\begin{cases} \pi_0 = \frac{1}{3} \pi_1 \\ \pi_1 = \pi_0 + \pi_2 \\ \pi_2 = \frac{2}{3} \pi_1 \end{cases}$$

得 $(\frac{1}{6}, \frac{1}{2}, \frac{1}{3})$

例 $\mu_0 = 6, \mu_1 = 2, \mu_2 = 3$.

$d(1) = d(0) = d(2) = 2$.

12) $P^2 = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & 1 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} \end{pmatrix}$



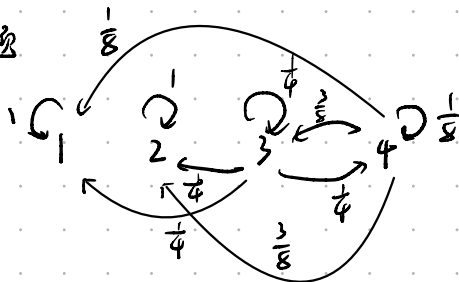
$\{1\}$ 闭

$\{0, 2\}$ 闭

$d(1) = 1 = d(0) = d(2)$

$\mu_1 = 1, \mu_0 = 3, \mu_2 = \frac{3}{2}$

19. 由题

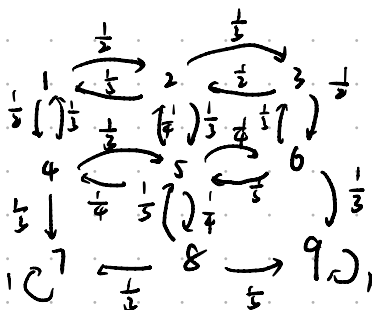


1 为吸收态. 设 h_i 为从 $x_0 = i$ 出发在有限时间到达 1 的概率

$$\begin{cases} h_1 = 1 \\ h_2 = 0 \\ h_3 = \frac{1}{4}h_3 + \frac{1}{4}h_2 + \frac{1}{4}h_1 + \frac{1}{4}h_4 \\ h_4 = \frac{1}{8}h_4 + \frac{1}{8}h_1 + \frac{3}{8}h_2 + \frac{3}{8}h_3 \end{cases}$$

则 $P\{T_1 < \infty | x_0 = 3\} = h_3 = \frac{4}{9}$

20. 由题



$I = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$$\begin{pmatrix} 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{3} & 0 & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{5} & 0 & \frac{1}{5} & 0 & 0 & 0 & \frac{1}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & \frac{1}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

解: $x_0 = 1$ 设 h_i 为从 i 出发到 7 的概率.

$$\begin{cases} h_7 = 1 & h_9 = 0 \\ h_1 = \frac{1}{5}h_2 + \frac{1}{5}h_4 & h_2 = \frac{1}{5}h_1 + \frac{1}{5}h_3 + \frac{1}{5}h_5 \\ h_3 = \frac{1}{5}h_2 + \frac{1}{5}h_6 & h_4 = \frac{1}{5}h_1 + \frac{1}{5}h_5 + \frac{1}{5}h_7 & h_5 = \frac{1}{4}(h_1 + h_4 + h_6 + h_8) \\ h_6 = \frac{1}{3}(h_3 + h_5 + h_9) & h_8 = \frac{1}{5}(h_5 + h_1 + h_9) \end{cases}$$

则 $h_1 = \frac{3}{7}$

21. (1) 在未相遇时, 两者独立运动, 由例 1.11 可知都是时齐马尔可夫链.

易知 Ω 也是.

$$\Omega = \{(0,0), (0,1), (1,0), (1,1)\}$$

则一步转移矩阵

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0.1 & 0.4 & 0.1 & 0.4 \\ 0.2 & 0.3 & 0.2 & 0.3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(2) 设从 i 状态出发到 $(0,0)$ 相遇的概率为 h_i

$$\begin{array}{cccc} (0,0) & (0,1) & (1,0) & (1,1) \\ h_1 & h_2 & h_3 & h_4 \end{array}$$

$$\begin{cases} h_1 = 1 & h_4 = 0 \\ h_2 = 0.1h_1 + 0.4h_2 + 0.1h_3 + 0.4h_4 \\ h_3 = 0.2h_1 + 0.3h_2 + 0.2h_3 + 0.3h_4 \end{cases}$$

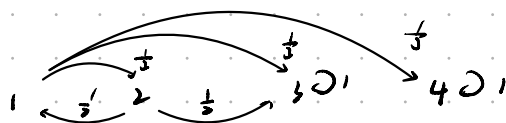
$$P = h_2 = \frac{2}{9}$$

(3) $a_1 \quad a_2 \quad a_3 \quad a_4$

$$\begin{cases} a_1 = 0 \\ a_2 = 1 + 0.1a_1 + 0.4a_2 + 0.1a_3 \\ a_3 = 1 + 0.2a_1 + 0.3a_2 + 0.2a_3 \end{cases}$$

则 $a_2 = 2 = a_3$

22.



解: (1) $P = \begin{cases} (\frac{1}{2} \frac{1}{2})^{\frac{n}{2}}, & n \text{ 为偶} \\ \frac{1}{2} \frac{1}{2} (\frac{1}{2} \frac{1}{2})^{\frac{n-1}{2}} + \frac{1}{2} \frac{1}{2} (\frac{1}{2} \frac{1}{2})^{\frac{n-1}{2}}, & n \text{ 为奇} \end{cases}$

$$\begin{cases} h_4 = 1 & h_3 = 0 \\ h_1 = \frac{1}{2}h_2 + \frac{1}{2}h_3 + \frac{1}{2}h_4 \\ h_2 = \frac{1}{2}h_1 + \frac{1}{2}h_3 \end{cases}$$

$$P = \frac{1}{2}h_1 + \frac{1}{2}h_2 = \frac{2}{10}$$

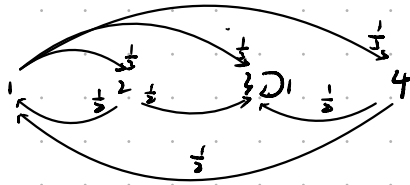
(2) $E = \frac{1}{10} \times 5 + \frac{7}{10} \times 1 = 2.2$

14) 将 3, 4 视为一个状态.

$$\begin{cases} a_1 = 1 + \frac{1}{3}a_2 + \frac{2}{3}a_3 \\ a_2 = 1 + \frac{1}{3}a_1 + \frac{1}{3}a_3 \\ a_3 = 1 \end{cases}$$

$$E(T) = \frac{1}{3}(a_1 + a_2) = 2$$

23.



$$1) P(X_1=4, X_2=2) = \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{27}$$

$$2) P(X_2=1 | X_1=3) = P(X_1=3 | X_2=1) \cdot \frac{P(X_1=1)}{P(X_1=3)} = \frac{1}{3} \times \frac{\frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3}}{1 - (\frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3})} = \frac{1}{7}$$

$$3) P = \begin{cases} (\frac{1}{3} \times \frac{1}{3})^{\frac{n-1}{2}} \cdot \frac{1}{3} & n \text{ 为奇数} \\ 0 & n \text{ 为偶数} \end{cases}$$

$$14) P(T_2 < T_3 | X_0=1) = h_1$$

$$\begin{cases} h_1 = \frac{1}{3}h_2 + \frac{1}{3}h_3 + \frac{1}{3}h_4 \\ h_2 = 1 \\ h_3 = 0 \\ h_4 = \frac{1}{3}h_1 + \frac{1}{3}h_5 \end{cases}$$

$$h_1 = \frac{2}{5}$$

$$15) \begin{cases} a_1 = 1 + \frac{1}{3}a_2 + \frac{1}{3}a_3 + \frac{1}{3}a_4 \\ a_2 = \frac{1}{3}a_1 + \frac{1}{3}a_3 + 1 \\ a_3 = 0 \\ a_4 = \frac{1}{3}a_1 + \frac{1}{3}a_3 \end{cases}$$

$$a_1 = \frac{5}{2}$$